NAGARSETU • नगरसेतू

GOVERNMENT OF MAHARASHTRA • PUNE MUNICIPAL CORPORATION (PMC)

NAGARSETU (नगरसेतू) — COMPLETE COMPREHENSIVE OPERATING MANUAL & FEATURE GUIDE

Pune Municipal Grievance Redressal & Server-Enforced Workflow Engine

Government of Maharashtra • Pune Municipal Corporation (PMC)

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1. EXECUTIVE SUMMARY & ARCHITECTURE

NagarSetu (नगरसेतू) is a modern, transparent municipal grievance resolution platform designed specifically for the Pune Municipal Corporation (PMC). It replaces legacy, opaque complaint boxes with an immutable, server-enforced civic engine.

Core Architectural Pillars

1. **Server-Enforced State Machine**: Status transitions cannot be skipped by officers or staff. Every state transition follows rigid business rules guarded by Role-Based Access Control (RBAC).
2. **250m Spatial Deduplication**: If multiple citizens report the same civic issue (e.g., a pothole or water main leak) within a 250-metre radius, the engine automatically clusters them into a single physical **Issue**, preventing administrative queue congestion while automatically escalating priority.
3. **Dual Client & Server EXIF Forensic Extraction**: Camera GPS coordinates (latitude, longitude) and timestamps are parsed directly in the citizen's browser and re-extracted from raw image binary on the server to prevent GPS spoofing.
4. **Statutory SLA Deadlines & Background Escalation**: Every complaint has statutory response and resolution deadlines. An automated cron background monitor escalates overdue tickets to senior supervisors and raises priority to `CRITICAL`.
5. **Citizen Verification & 7-Day Reopening Window**: When an officer marks a complaint resolved, it enters `RESOLVED_AWAITING_VERIFICATION`. The original citizen has a statutory 7-day window to inspect the

resolution (with photo evidence) and either confirm closure or reopen the ticket.

6. **Immutable Audit Trail (`AuditEvent`)**: Every accepted state transition, supervisor deadline modification, and unauthorized attempt is permanently logged into an append-only audit database table.

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2. SYSTEM ROLES & ACCESS CONTROL MATRIX

The platform provides 4 distinct user roles:

Role	Access Scope	Allowed Actions
CITIZEN	Public Portal (`/`, `/report`, `/track`, `/my-complaints`, `/map`, `/scorecard`)	Lodge grievances, upload photos, track tickets by reference token, confirm resolution, reopen issues within 7 days.
OFFICER	Staff Portal (`/staff`), Assigned Department Issues	View assigned issues, change status to `IN_PROGRESS`, add progress notes, submit resolution notes and "after" photo proof. Cannot reassign or extend deadlines.
SUPERVISOR	Staff Portal (`/staff`), Department Queue & Overdue Queue	Reassign issues to officers, modify statutory SLA deadlines with mandatory audit reasons, monitor escalated cases.
ADMIN	Full Platform (`/staff`, Internal Engine Console)	Department-wide oversight, manage SLA and priority thresholds, inspect immutable audit logs, view cross-department matrices.

Pre-Configured 1-Click Demo Accounts

For rapid testing and evaluation, the login page (`/login`) and top navigation bar include instant 1-click accounts:

- **Citizen**: `aarav.sharma@citizen.in` (Password: `PuneDemo@2026`)
- **Citizen**: `sneha.apte@citizen.in` (Password: `PuneDemo@2026`)
- **Officer (Roads)**: `officer.roads@pune.gov.in` (Password: `PuneDemo@2026`)
- **Officer (Water)**: `officer.water@pune.gov.in` (Password: `PuneDemo@2026`)
- **Supervisor (West Ward)**: `supervisor.west@pune.gov.in` (Password: `PuneDemo@2026`)
- **PMC Admin**: `admin@pune.gov.in` (Password: `PuneDemo@2026`)

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3. CITIZEN GRIEVANCE REGISTRATION STEP-BY-STEP (`/report`)

The citizen intake flow is a resilient 7-step wizard that functions with or without AI assistance:

Step 1: Category Selection

- Choose from 5 statutory municipal categories:
- `[RD] Roads & Infrastructure` (Potholes, broken footpaths, traffic signals, streetlights)
- `[WT] Water & Sewage` (Pipeline bursts, contaminated water, open drains, low pressure)
- `[SW] Waste Management & Sanitation` (Garbage dumps, missed collection, dead animals)
- `[SF] Civic Nuisance & Public Safety` (Illegal hawking, encroachments, noise, construction debris)
- `[GN] Other Municipal Services` (Tree trimming, stray animals, park maintenance)
- Select the applicable subcategory. If accessing via `?category=roads-infrastructure`, the category is automatically preselected.

Step 2: Problem Description

- Enter the detailed description in the citizen's own words.
- **Multilingual Guarantee**: Citizen text is preserved verbatim in Hindi, Marathi, or English. The backend never rewrites or auto-translates citizen complaint text.
- Voice-to-Text: Supports speech recognition via the microphone button.

Step 3: Photographic Evidence & Media Upload

- Take a photo directly using the device camera (`capture="environment"`) or select a gallery file.
- Supported formats: JPEG, PNG, WebP (up to 10 MB); MP4 (up to 30 MB).
- Client and server enforce strict MIME type and file signature validation. Uploads are stored securely outside the `/public` root.

Step 4: Automatic GPS Location & Forensic EXIF Extraction

- The browser parses the photograph's binary metadata using `exifr`.
- If GPS coordinates exist (`latitude`, `longitude`, `DateTimeOriginal`):
- Automatically pins the exact incident spot on the Pune OpenStreetMap.
- If the device GPS differs by >100m from the photo EXIF, the system displays an alert with a single-click button to choose either the photo location or current device position.
- If no EXIF data exists:
- Interactive map picker opens with Pune landmarks (Shivajinagar, Kothrud, Swargate, Hadapsar, Viman Nagar, Hinjawadi).
- Click anywhere in Pune to pin the location.

Step 5: Duplicate Detection Alert & Smart Merging

- If a complaint exists within **250 metres** under the same category:
- The system highlights the existing ticket reference.
- Citizens are offered 4 options:
    1. Track existing complaint.
    2. Attach their photo/evidence to the existing issue (elevating its priority).
    3. Modify description/location.
    4. Submit as a separate distinct grievance.

Step 6: Review & Final Submission

- Review all collected metadata, category, location, and photos.
- Submit ticket. The backend generates a permanent reference token (e.g., `PUNE-RD-884210`).

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4. CITIZEN TRACKING, VERIFICATION & REOPENING (`/track` & `/my-complaints`)

Public Tracking vs. Authenticated Citizen Tracking

- **Public View**: Anyone entering a tracking ID sees public status, anonymised location (~100m grid snap), department name, SLA countdown, and resolution timeline. Citizen personal info is strictly withheld.
- **Citizen Owner View**: When logged in as the reporting citizen, displays full raw evidence, detailed officer notes, and direct action buttons.

Resolution Verification Flow

1. When status reaches `RESOLVED_AWAITING_VERIFICATION`:
2. The citizen inspects the officer's resolution notes and mandatory "After" photograph.
3. **Confirm Closure**: Citizen clicks "Verify & Confirm Closure" → Ticket enters `CLOSED`.
4. **Reopen Ticket (7-Day Statutory Window)**:
5. If work is incomplete, citizen enters a reason and clicks "Reopen Complaint".
6. The ticket transitions back to `IN_PROGRESS` or `ASSIGNED` with escalated supervisor notification.

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5. INTERACTIVE LIVE GIS MAP (`/map` & Homepage)

- **OpenStreetMap + Leaflet Engine**: Real-time visualization centred on Pune (`18.5204°N, 73.8567°E`).
- **Privacy Grid Snapping**: Public map endpoints snap coordinates to a ~100m grid to protect citizen home

privacy.

- **Urgency & Priority Rings**:
- Green: `NORMAL` priority.
- Orange: `HIGH` priority.
- Red: `CRITICAL` priority.
- Pulsing Halo: Overdue SLA breach.
- **Department Badges**: `[RD] Roads`, `[WT] Water`, `[SW] Waste`, `[SF] Safety`, `[GN] General`.
- **Live Server-Sent Events (SSE)**: Map updates automatically when complaints are logged or updated without requiring page refresh.

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6. STAFF WORKFLOW PORTAL (`/staff`)

Officer Workspace

- Filtered table displaying issues assigned to the logged-in officer.
- Sorted by nearest SLA deadline.
- Allowed Transitions Engine (`GET /api/issues/[id]/allowed-transitions`): Officers only see buttons permitted for their role:
- `SUBMITTED` → `TRIAGED`
- `TRIAGED` → `ASSIGNED`
- `ASSIGNED` → `IN_PROGRESS`
- `IN_PROGRESS` → `RESOLVED_AWAITING_VERIFICATION` (Requires resolution note and photo proof)
- **AI Copilot Suggestions**: Drafts departmental response notes and verifies subcategory tagging.

Supervisor Workspace

- **Overdue & Escalation Queue**: Instant view of tickets where SLA has breached.
- **Reassignment**: Move tickets across ward officers with a required justification note.
- **Deadline Extension**: Adjust resolution deadlines with mandatory audit reasons; the original SLA deadline is permanently preserved for accountability reporting.

Scenario H (Tamper & Refusal Testing)

- Evaluators can simulate illegal state skips (e.g., trying to jump directly from `SUBMITTED` → `CLOSED` without triage).
- The server immediately rejects the request with `403/422 REFUSED` and records an unauthorized attempt in the `AuditEvent` table.

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7. MUNICIPAL SCORECARD & PERFORMANCE METRICS (`/scorecard`)

Statutory performance metrics computed from the database: * **Resolution Rate (%)**: `(Resolved Issues ÷ Eligible Complaints) × 100` * **On-Time Resolution Rate (%)**: `(Resolved Before Deadline ÷ Eligible Resolved) × 100` * *Note: Invalid / Rejected tickets are excluded from calculations so numbers cannot be inflated.*

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8. AI ASSISTANT & BILINGUAL ENGINE

- **Floating Assistant**: Accessible on all pages for conversational complaint intake.
- **Languages**: English, Hindi (हिंदी), and Marathi (मराठी).
- Language selection persists across cookies and dynamically updates labels, navigation, and validation notices.

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9. STEP-BY-STEP CHEAT SHEET FOR EVALUATORS

1. **Open the Website**: Visit `http://10.60.244.171:3000` (or `http://localhost:3000`).
2. **Lodge a Complaint**:
3. Click "Register New Grievance".
4. Select `Roads & Infrastructure` → `Potholes`.
5. Use the demo test button: "Test Photo WITH Real GPS EXIF (Kothrud)".
6. Notice automatic coordinate detection (18.5074°N, 73.8077°E).
7. Click Submit and copy the generated reference ID (e.g. `PUNE-RD-...`).
8. **Log in as Officer**:
9. Click "Sign In" in the top bar.
10. Click "Officer Roads" 1-click login.
11. Go to Staff Portal → locate the newly lodged ticket.
12. Move to `IN_PROGRESS` → `RESOLVED_AWAITING_VERIFICATION`.
13. **Log in as Citizen to Verify**:
14. Log back in as `Aarav Sharma`.
15. Go to "My Complaints" or `/track/[referenceId]`.
16. Click "Verify & Confirm Closure" or test the 7-day reopening mechanism.
17. **Inspect Audit Trail**:
18. In Staff Portal or Evaluator Console, open the "Audit Trail" tab to review every accepted transition and refusal event.
